

Rohan Bahl

rohan_bahl@u.nus.edu | rb9823.github.io/website/ | linkedin.com/in/rohan-bahl | github.com/RB9823 | +65 91471451

WORK AUTHORIZATION

Singapore Citizen; eligible for **H-1B1** and **J-1 US visas**, and an **HPI UK visa, with no petition required**

EDUCATION

National University of Singapore Expected: Dec 2025
Bachelor of Computing (Hons), Computer Science (Specialisations: Artificial Intelligence & Database Systems)
• Special Programme: **NUS Overseas Colleges Toronto–Waterloo AY 24/25 Sem 2 Intake**

EXPERIENCE

The Tudor Group May 2025 – Aug 2025
Software Engineer Intern Singapore
• Built and shipped a horizontally scalable distributed real-time alerting platform (4 Spring Boot microservices, ActiveMQ, Redis, WebSocket) processing high-volume Grafana alerts with message deduplication and aggregation, delivering sub-second updates to the UI, replacing the legacy Nimsoft platform for the Site Reliability team
• Reduced Grafana infrastructure management time by 99% (10 hrs → 5 mins) by building an Ansible + Python automation framework using the Grafana REST API for dashboard migration, alert generation, and multi-environment config management across dev, staging, and prod
• Cut trade log extraction time by 95% (60s → 3s) and eliminated legacy licensing costs by delivering a FIX log viewer (React, TypeScript, TanStack Table) in one sprint, integrated with Spring Boot APIs for the global TradeOps team

Upside Robotics Aug 2024 – Mar 2025
Software Engineer Intern Waterloo
• Architected and deployed a serverless event-driven telemetry pipeline (Lambda, S3, DynamoDB) via IaC (Serverless Framework), validating and processing 5M+ sensor records per run (including image compression/classification) into structured time-series data for real-time fleet monitoring across 5+ autonomous robots
• Reduced manual field diagnostics by 30% by building a real-time operations dashboard (Next.js, TypeScript, Mapbox) with live telemetry visualization and geospatial tracking
• Ensured reliable data delivery over intermittent LTE networks by collaborating with hardware engineers to implement resilient firmware logic (batching + offline caching)

PROJECTS

StoryForge (Cursor Hackathon 2025) Oct 2025 – Nov 2025
ML Engineer / Team Lead Singapore
Built a 2D AI-powered game engine that generates infinite, procedurally generated worlds with consistent narratives and visuals using Google Gemini AI, fal.ai, dynamic audio using Elevenlabs voice agents
• Finalist, Top 4% (15 / 400 projects)

GUI Murphy (TikTok TechJam 2025) Aug 2025 – Sept 2025
ML Engineer / Team Lead Singapore
Developed a multi-stage UI identification and consistency validation pipeline leveraging OmniParser, CLIP, and MLLMs
• Finalist, Top 4% (12 / 300 projects)

Smart Storybook, (ETHGlobal SF 2024) Oct 2024 – Oct 2024
Backend Engineer San Francisco
A Web3 platform enabling creators to monetize intellectual property through NFTs
• Built a FastAPI backend integrating Pinata’s IPFS API for decentralized asset storage; Won 2nd place, Best AI Application

SKILLS

- **Programming Languages:** Python, Java, JavaScript/TypeScript, C
- **Databases:** MongoDB, MySQL, PostgreSQL, ChromaDB, DynamoDB
- **Frameworks & Libraries:** Spring Boot, FastAPI, Flask, React, Next.js, Streamlit, Serverless Framework, PyTorch, Ollama
- **DevOps & Tools:** Git, GitHub Actions, Ansible, Redis, ActiveMQ, Grafana, Hadoop (HDFS & MapReduce), Spark, Flink
- **Cloud & Infrastructure:** AWS (EC2, S3, Lambda, Cognito, CloudWatch), Vercel